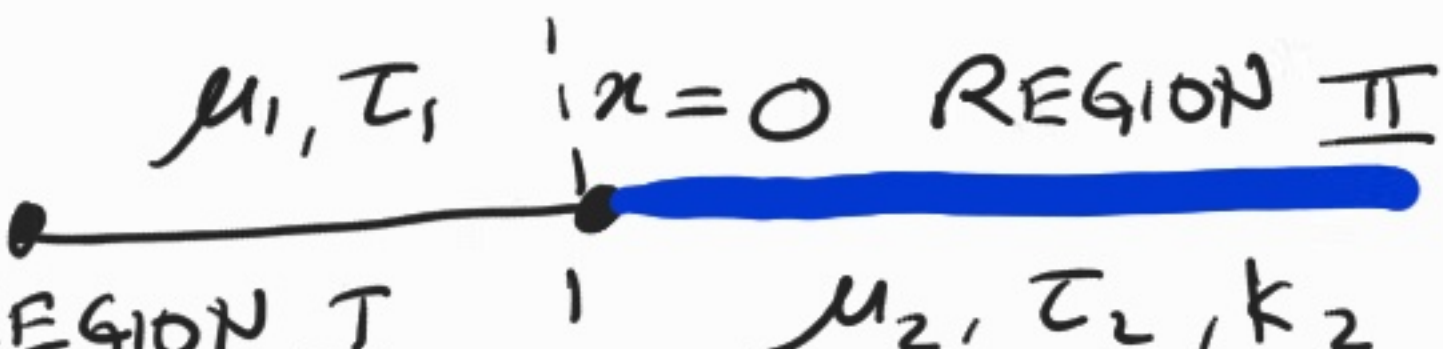
Reflection / Transmission at Boundary between 2 media

- **Boundary Condition Application** → leads to the equations for the amplitudes of the reflected & transmitted waves.



⇒ **Incident Wave** :

$$y_i = y_{m,i} \sin(k_1 x - \omega t)$$

⇒ **Reflected Wave** :

$$y_r = y_{m,r} \sin(k_1 x + \omega t)$$

⇒ **Transmitted Wave** :

$$y_t = y_{m,t} \sin(k_2 x - \omega t)$$

Transmission of Waves at Boundaries

- **Boundary Conditions :**

$$y_i + y_r = y_t \quad \text{at } x = 0$$

→ forward moving.

$$y_i = (y_m)_i \sin(\cancel{k_1 x}^{\,0} - \omega t)$$

→ backward moing

$$y_r = (y_m)_r \sin(\cancel{k_1 x}^{\,0} + \omega t)$$

→ forward moing.

$$y_t = (y_m)_t \sin(\cancel{k_2 x}^{\,0} - \omega t)$$

$$\therefore (y_m)_i \sin(-\omega t) + (y_m)_r \sin(\omega t)$$

$$\therefore = (y_m)_t \sin(-\omega t)$$

next page →

$$\Rightarrow \underbrace{-y_{m,i} + y_{m,r} = -y_{m,t}}$$

Amplitude Equivalence

$$\therefore \boxed{y_{m,i} = y_{m,r} + y_{m,t}}$$

At $x = 0$, since

$y_i + y_r = y_t$, then

$$\boxed{\frac{\partial y_i}{\partial x} + \frac{\partial y_r}{\partial x} = \frac{\partial y_t}{\partial x}}$$

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Impedance definition (string) :

$$Z = \frac{\text{Force on a string } (F_y)}{\text{Velocity Amplitude } (v_y)}$$

$$y = y_m \sin(kx - \omega t)$$

$$F_y = \tau(\partial y / \partial x) \quad \text{——— (1)}$$

$$v_y = \partial y / \partial t \quad \text{——— (2)}$$

$$\boxed{Z = \frac{\tau k}{\omega}} = \frac{\tau}{v} \quad \left| \text{since } v = \frac{\omega}{k} \right.$$

$$Z = \frac{\tau}{v} = \sqrt{\tau \mu} \quad (\text{Impedance})$$

Returning to the differential Boundary Condition:

$$\frac{\partial y_i}{\partial x} + \frac{\partial y_r}{\partial x} = \frac{\partial y_t}{\partial x}$$

$$k_1 y_{m,i} \cos(\cancel{kx}^{0} - \omega t) + k_1 y_{m,r} \cos(\cancel{kx}^{0} + \omega t) \quad \text{— (3)}$$

$$= k_2 y_{m,t} \cos(\cancel{kx}^{0} - \omega t)$$

Since $Z_1 = (\tau k / \omega)$

$$k_1 = \frac{Z_1 \omega}{\tau_1} \quad \& \quad k_2 = \frac{Z_2 \omega}{\tau_2}$$

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$$k_1 y_{m,i} + k_1 y_{m,r} = k_2 y_{m,t}$$

$$y_{m,r} = \left[\frac{Z_2 - Z_1}{Z_1 + Z_2}\right] y_{m,i}$$

$$y_{m,t} = \left[\frac{2Z_1}{Z_1 + Z_2}\right] y_{m,i}$$

Reflection / Transmission at Boundary between 2 media

- Applying boundary conditions leads to equations for the amplitudes of the reflected & transmitted waves.

⇒ **Reflection Coefficient**

$$r = \frac{\text{Amplitude of Reflected wave}}{\text{Amplitude of incident wave}}$$

$$= \frac{y_{m,r}}{y_{m,i}} = \left[\frac{Z_2 - Z_1}{Z_1 + Z_2}\right]$$

⇒ Transmission Coefficient.

$$t = \frac{\text{Amplitude of transmitted wave}}{\text{Amplitude of incident wave}}$$

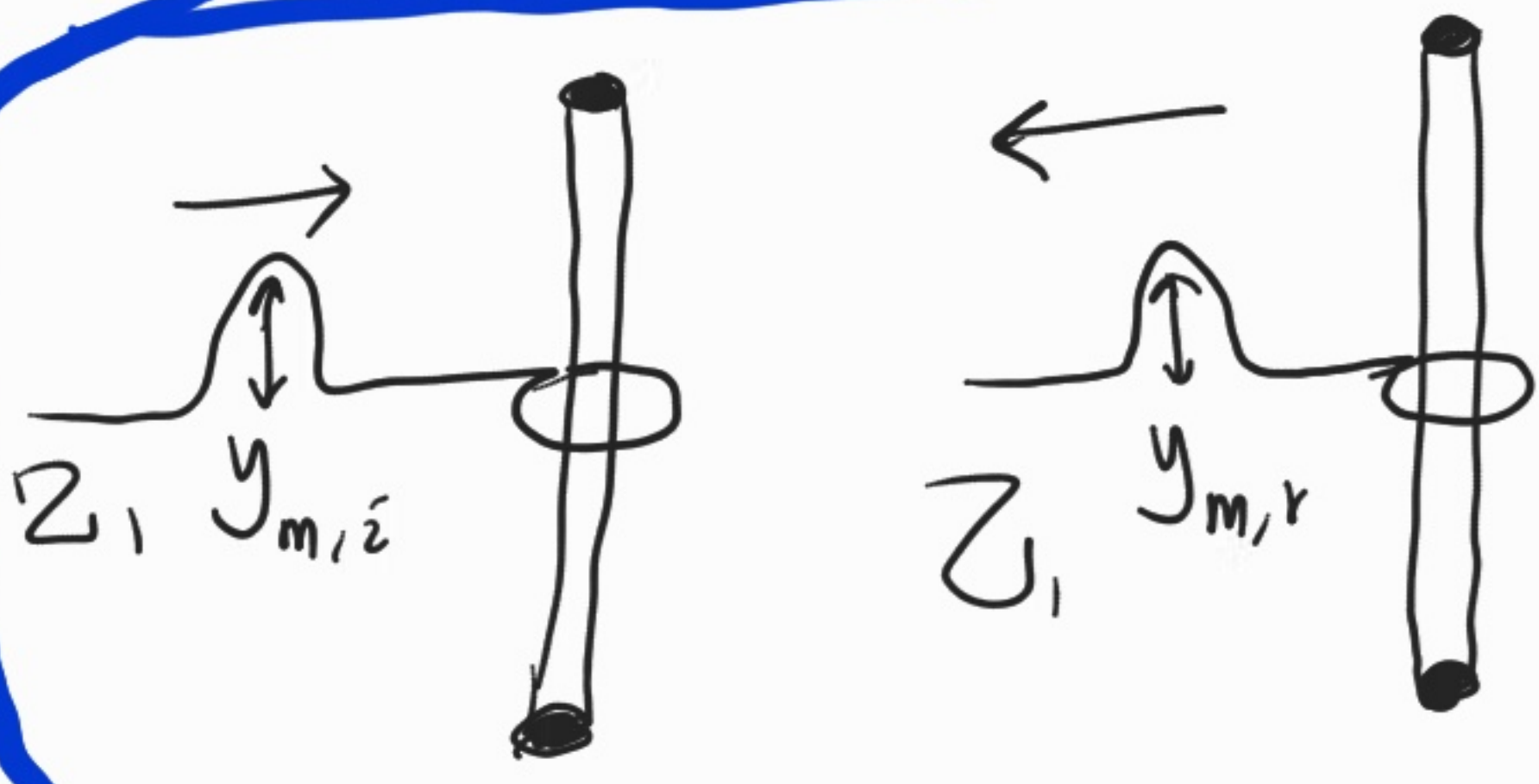
$$= \frac{y_{m,t}}{y_{m,i}} = \left[\frac{2Z_1}{Z_1 + Z_2}\right]$$

Reflection / Transmission at Boundary between 2 media.

⇒ Special Case for coefficients t & r:

$Z_2 = 0$

⇒ This string is attached to a ring that can move up & down on a vertical rod.



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then the coefficients:

$$r = \frac{Z_2 - Z_1}{Z_1 + Z_2}$$

$\Rightarrow$ Allow for rod-ring motion (same amplitude sign)

$$r = \frac{-(\cancel{Z_2}^{0} - Z_1)}{Z_1 + \cancel{Z_2}^{0}} = \frac{Z_1}{Z_1} = 1$$

$$\therefore \; y_{m,r} = y_{m,i}$$

$$t = \frac{(y_{m,t} \overset{\#}{=} 0)}{y_{m,i}} = 0 \left.\right\} \text{No transmitted wave}$$

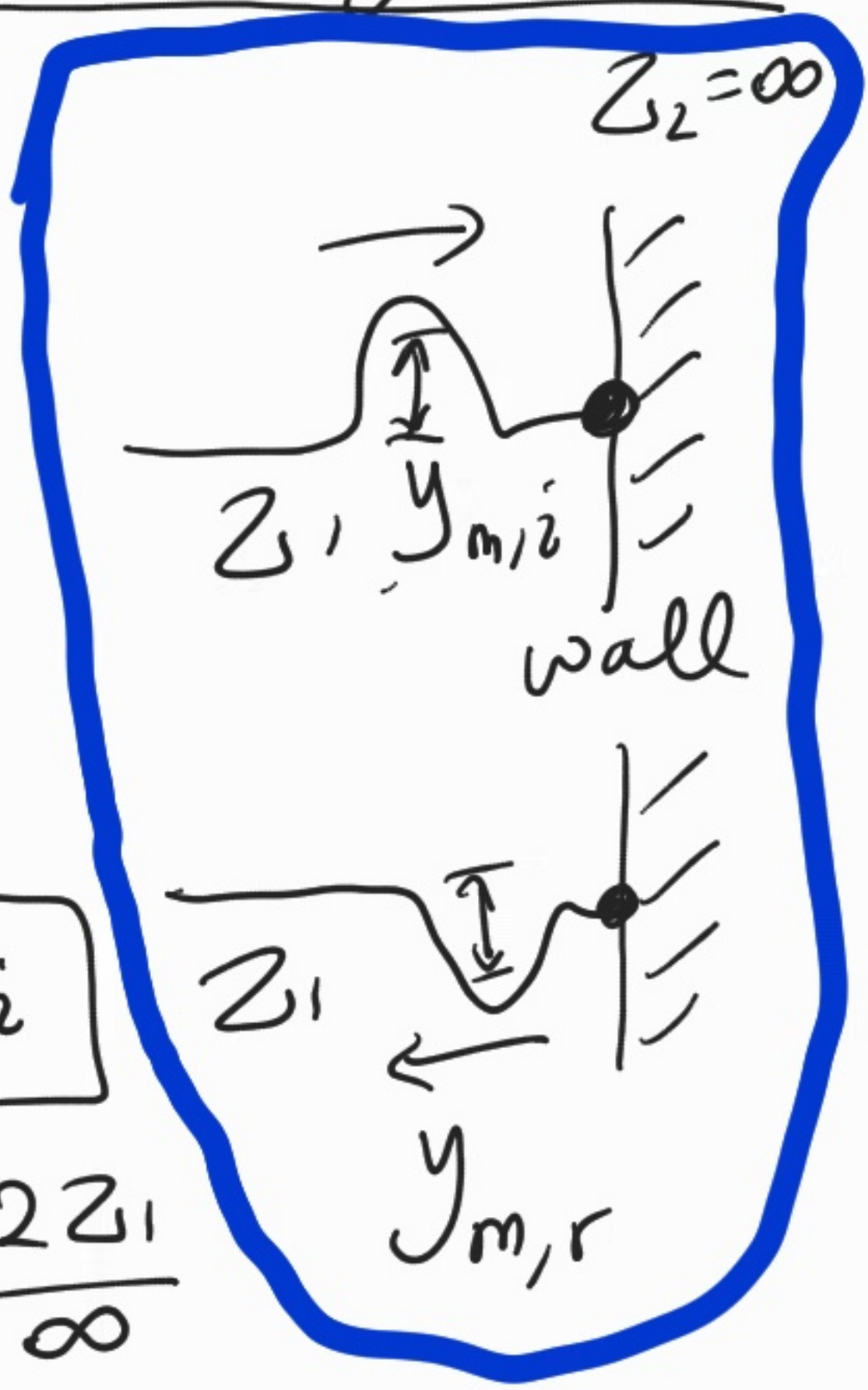
Special Case for String attached to wall

$$r = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

$$= \frac{\infty - Z_1}{\infty + Z_1}$$

$$r \simeq -1$$

$$\boxed{y_{m,r} = -y_{m,i}}$$



$$t = \frac{2Z_1}{Z_1 + Z_2} = \frac{2Z_1}{\infty}$$

$$t \approx 0$$

- No transmitted wave trough wall.

Power Reflection & Transmission

- Average Power transported along the string in terms of Z:

$$P_{ave} = \frac{1}{2}\mu v y_m^2 \omega^2 = \frac{1}{2} Z y_m^2 \omega^2$$

$$Z = \sqrt{\mu\tau} = \sqrt{\mu^2 v^2} \rightarrow Z = \mu v$$

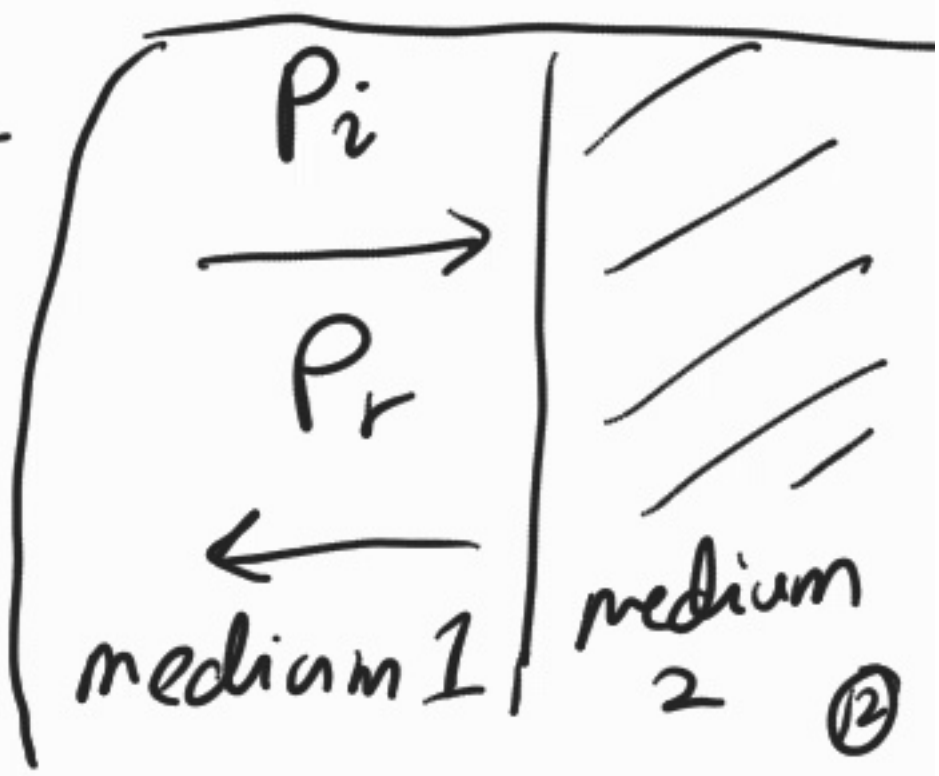
$$v = \sqrt{\tau/\mu} \longrightarrow \tau = \mu v^2$$

- power Reflection Coefficient.

$$R = \frac{P_r}{P_i} = \frac{\frac{1}{2} Z_1 y_{m,r}^2 \omega^2}{\frac{1}{2} Z_1 y_{m,i}^2 \omega^2}$$

$$R = \frac{y_{m,r}^2}{y_{m,i}^2} = r^2$$

#



- The reflected & incident power travel in the same medium with impedance, Z_1.
- Power Transmission Coefficient

$$T = \frac{P_t}{P_i} = \frac{\frac{1}{2} Z_2 y_{m,t}^2 \omega^2}{\frac{1}{2} Z_2 y_{m,i}^2 \omega^2} = \frac{Z_2}{Z_1} \left[\frac{y_{m,t}}{y_{m,i}} \right]^2$$

$$T = \frac{Z_2}{Z_1} t^2 \quad \#$$

- Transmitted wave travels in the medium with Z_{12}.

